System Design: Introduction & Types

Easy English notes from your Hindi YouTube lecture (about 2 minutes).

1. What is System Design?

System design is the process of planning the different parts of a software application and deciding how those parts work together.

Imagine you want to build a large shopping app. Before writing all the code, you plan its structure, its modules, and how information moves between them.

2. What do we plan?

• **Architecture:** the overall structure of the application.

• **Modules:** separate parts, such as login, products, orders and payments.

• **Interactions:** how the modules communicate.

• **Data flow:** how information travels from one part to another.

3. Two levels of system design

The transcript becomes unclear while naming the two types. The explanation appears to refer to **High-Level Design (HLD)** and **Low-Level Design (LLD)**. The following labels and examples are provided to clarify that likely meaning.

High-Level Design (HLD)	Low-Level Design (LLD)
Big picture of the system	Detailed design inside each component
Major modules, architecture and communication	Classes, methods, business logic and implementation details
Example: App talks to API, database and payment service	Example: How OrderService validates and saves an order

4. Simple example: Online shopping app

HLD: Customer App → Backend API → Order Service → Database. A Payment Service handles payments.

LLD: An OrderService class has methods such as createOrder(), calculateTotal() and saveOrder(). These methods contain the actual business rules.

5. One-minute revision

System design: Plan the parts of a software system and how they connect.

HLD: What are the major components, and how do they communicate?

LLD: How will each component be built in detail?

6. Interview practice

Q1. What is system design? It is planning the architecture, modules, interactions and data flow of a software system.

Q2. What is the difference between HLD and LLD? HLD shows the overall structure; LLD describes internal design and implementation details.

Q3. Why do we need system design? It helps developers understand the system before implementation and coordinate work across components.

Source note: The supplied auto-generated Hindi transcript contains several unclear or missing phrases, especially around 0:50–2:00. These notes summarize the clear points and mark the likely HLD/LLD interpretation; they are not a verbatim transcript.